Sahaj Bhadja

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Summary

Software Engineer with 2+ years building tested, object-oriented software in **C++**, **Python**, and Java, with **robotics** motion-planning research, user-facing applications, **unit testing** and code coverage, and Agile Scrum delivery on Linux and Docker.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant asynchronous job processing in Python with **explicit state transitions** and failure isolation, processing 1K+ jobs at 3x throughput without cascading failures under concurrent production load.
- Shipped integration pipelines through a **CI/CD-backed release** process with data validation and reconciliation checks, sustaining 99%+ reliability across 5K+ monthly jobs, and tuned indexed queries cutting dashboard latency from 1.2s to 350ms.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a **React** user interface and Flask backend for upload, rubric, and grading workflows, serving 150+ instructors and 10K+ daily requests on AWS while cutting feedback turnaround 60%.
- Integrated **image processing** with GPT-4o Vision OCR and Claude Vision fallback for handwritten answers, plus multithreaded batch execution, achieving 90%+ grading consistency across 10K+ text and handwritten submissions.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a Decision Transformer **motion planner** for autonomous driving, transforming 100K+ trajectory logs and nearby-agent states into a **global coordinate frame**, improving long-horizon trajectory planning 18% in closed-loop simulation.
- Designed a spatial-temporal encoder fusing **HD-map geometry**, lane direction, and road boundaries with ego trajectories via distance-aware attention, improving closed-loop planning stability 15% and reducing off-road trajectories.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built **object-oriented** modules using Singleton and Factory patterns and automated 20+ unit and integration suites with PyTest and Mockito, publishing **code coverage** summaries and failure logs from nightly Jenkins runs.
- Validated cross-platform executables on Windows, Linux, and macOS VMs across feature, staging, and production branches in **Agile Scrum**, and optimized APIs to improve response times 30% while reducing technical debt 20%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a React and Node.js resource-management platform with RESTful microservices and real-time WebSocket updates, syncing 2M+ records across locations and supporting **code review** and rollout across 6+ core workflows.
- Ran Jenkins pipelines with linting, **static analysis**, Snyk and OWASP dependency checks, and unit, integration, and end-to-end tests before **Docker** packaging, reaching 95%+ successful build validation across release runs.

Skills

Languages: C++, Python, Java, JavaScript, TypeScript, SQL, Bash
Software Engineering: Object-Oriented Design, Design Patterns, Data Structures, Algorithms, SDLC, Code Reviews, Debugging
Testing & Quality: Unit Testing, Integration Testing, Code Coverage, Static Analysis, PyTest, Mockito, Jest, Selenium, Snyk
Robotics & Vision: Motion Planning, Simulation, Coordinate Transforms, PyTorch, Transformers, OCR, Image Processing
Tools & Platforms: Linux, Docker, Git, Jenkins, CI/CD, GitHub Actions, React, Agile/Scrum, Windows, macOS

Projects

3D Implant Planning Viewer | *C++, Qt/QML, OpenGL, Eigen, CMake, GoogleTest*

- Designed a **C++ and Qt/QML** desktop application that loads 3D bone meshes, lets users position implant models through rigid-body transforms, and computes alignment angles with **linear algebra** in unit-tested geometry modules.

Robotic Arm Boundary Monitor | *C++, Eigen, Linux, Docker, GoogleTest, Valgrind*

- Designed a C++ service computing robot tool-tip pose through forward kinematics and enforcing **virtual boundary** constraints continuously, verified with GoogleTest, code coverage, clang-tidy **static analysis**, and Valgrind memory checks.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2, OS Scheduling*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel queue scheduling in **C++** with React controls and Gantt charts, testing 1K+ process combinations and cutting average turnaround time 25%.